

LB-UNet++: LBUNet with Attention Fusion Skip Connection for Skin Lesion Segmentation

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Abstract—Background:

Deep learning models, particularly U-Net and its variants, have demonstrated strong performance in medical image segmentation tasks. Among them, LBUNet introduces boundary-aware supervision and group-shuffle blocks to improve segmentation accuracy. However, its architecture remains computationally heavy due to multiple fusion branches and auxiliary outputs.

Objective:

This study aims to simplify the LBUNet architecture by introducing a learnable attention-based skip connection mechanism—AttentionSkipFusion (ASF)—to replace hand-crafted gated fusion, and by removing auxiliary outputs to reduce model complexity and improve inference efficiency.

Methods:

We designed a modified version of LBUNet incorporating ASF modules in all decoder stages and replaced the complex fusion and keypoint branches with standard convolution blocks. Both the baseline and modified models were trained on the same dermoscopic skin lesion segmentation dataset under identical experimental settings. Key metrics such as mean Intersection over Union (mIoU) and Dice Similarity Coefficient (DSC) were used for evaluation.

Results:

The modified LBUNet achieved approximately 40% faster inference speed compared to the baseline model. However, this came with a slight reduction in segmentation performance: mIoU dropped from 0.8111 to 0.8043 and DSC from 0.8899 to 0.8824 on the test set. Validation results showed a similar pattern.

Conclusion:

While ASF offers architectural simplicity and runtime efficiency, its integration alone is insufficient to surpass the performance of the original LBUNet, primarily due to the removal of deep supervision. The results suggest that attention-based fusion should be complemented by auxiliary learning signals to retain performance advantages. Future work may explore hybrid fusion mechanisms, transformer-based encoders, or knowledge distillation from deeper supervised models.

Index Terms—Medical Image Segmentation, LBUNet, AttentionSkipFusion, Skin Lesion Analysis, Deep Learning, U-Net, Boundary-Aware Supervision, Semantic Segmentation, Lightweight Networks, Dermoscopic Imaging

I. INTRODUCTION

Skin cancer is among the most prevalent forms of cancer globally, with melanoma accounting for a significant proportion of skin cancer-related fatalities [9]. Early and accurate segmentation of skin lesions from medical images such as dermoscopy photographs is crucial for timely diagnosis and effective treatment planning [10]. However, manual seg-

mentation by dermatologists is often labor-intensive, time-consuming, and subject to inter- and intra-observer variability. Consequently, there has been an increasing interest in developing automated segmentation techniques that leverage advancements in artificial intelligence and deep learning to assist clinicians in more accurate and efficient diagnostics [11].

Deep learning-based methods, particularly convolutional neural networks (CNNs), have significantly advanced the state-of-the-art in medical image segmentation tasks. Among these, the U-Net architecture [6] has become a seminal framework, known for its encoder-decoder structure and skip connections that effectively preserve spatial information critical for accurate boundary delineation. Nevertheless, despite its effectiveness, U-Net-based models typically face challenges related to handling ambiguous lesion boundaries, limited global contextual information, and high computational requirements, particularly when deployed on resource-constrained edge devices.

To address these issues, several recent approaches have been proposed to improve both segmentation accuracy and computational efficiency. Models such as MALUNet [1] and EGE-UNet [2] have introduced multi-attention mechanisms to enhance segmentation capabilities while maintaining low computational complexity, thus enabling practical deployment in clinical and mobile environments. Additionally, boundary-aware methods like LB-UNet [3], Boundary-aware Transformers (BAT) [4], and Structure Boundary Preserving Segmentation frameworks [5] explicitly focus on accurately segmenting ambiguous boundaries through innovative modules, effectively integrating boundary-specific domain knowledge into their learning processes.

Motivated by these advancements and ongoing challenges, this research proposes an enhanced lightweight U-Net variant (improve_lbunet) designed explicitly for skin lesion segmentation tasks. Our model incorporates a novel Attention Skip Fusion module and optimized convolutional layers, aiming to improve feature fusion between encoder and decoder pathways, thus effectively handling ambiguous lesion boundaries. By carefully integrating these enhancements, we aim to provide an efficient and accurate solution suitable for real-world clinical deployment, even under strict computational constraints.

The remainder of this paper is organized as follows. Section II presents a detailed literature review of recent advances in medical image segmentation, with a focus on lightweight

architectures and attention-based models. Section III outlines the original LBUNet architecture, its key components, and associated limitations. Section IV describes the proposed enhancements, including the AttentionSkipFusion module and redesigned decoder structure. Section V details the experimental design, datasets, preprocessing techniques, and evaluation metrics. Section VI discusses the experiment setup and implementation configurations. Section VII provides a comparative analysis of the baseline and modified models, followed by discussions and ablation studies. Finally, Section VIII concludes the paper and suggests future research directions, while acknowledgments and additional resources are provided in Sections IX and the Appendix, respectively.

II. LITERATURE REVIEW

Medical image segmentation, particularly skin lesion segmentation, plays a vital role in diagnostic assistance and clinical treatment strategies. Deep learning techniques, especially those based on the U-Net architecture [6], have greatly advanced segmentation capabilities. U-Net’s symmetric encoder-decoder structure combined with skip connections effectively preserves spatial information, leading to improved segmentation precision. Extensions of U-Net, including UNet++ [7] and Attention U-Net [8], further refined these capabilities by employing sophisticated feature propagation and region-specific attention mechanisms.

Recently, emphasis has shifted toward developing lightweight models suitable for deployment in resource-limited environments, such as mobile and edge devices. MALUNet [1] introduced a multi-attention framework that incorporates Dilated Gated Attention (DGA), Inverted External Attention (IEA), Channel Attention Bridge (CAB), and Spatial Attention Bridge (SAB) modules. These modules collectively enhance segmentation accuracy by simultaneously capturing global and local context, significantly reducing model parameters and computational complexity.

Further advancing lightweight models, Ruan et al. proposed EGE-UNet [2], which leverages Group multi-axis Hadamard Product Attention (GHPA) and Group Aggregation Bridge (GAB) modules. These innovations facilitate effective extraction and integration of multi-scale and multi-perspective information, achieving high segmentation performance with remarkably low computational resources. Such models are particularly suited for practical clinical applications where computational efficiency is crucial.

Addressing boundary ambiguity, Xu and Tong introduced LB-UNet [3], a lightweight architecture explicitly designed for precise boundary segmentation of skin lesions. This model employs a novel Group Shuffle Attention (GSA) module for computational efficiency and integrates a Prediction Map Auxiliary (PMA) module that uses Genetic Algorithm-based boundary key-point generation. These components significantly enhance the model’s ability to handle ambiguous lesion boundaries, demonstrating state-of-the-art results with minimal computational overhead.

Transformer-based approaches have also emerged as powerful segmentation methodologies. Wang et al. [4] presented a Boundary-aware Transformer (BAT) that effectively combines global self-attention mechanisms with boundary-specific attention gates. This model addresses the challenges posed by ambiguous lesion boundaries by integrating boundary-aware attention modules, thereby effectively capturing both local details and global contextual information.

In a similar direction, Lee et al. [5] proposed the Structure Boundary Preserving Segmentation framework, which explicitly preserves structural boundary information via specialized modules such as the Boundary Preserving Block (BPB) and Shape Boundary-aware Evaluator (SBE). These modules incorporate domain-specific knowledge into segmentation networks, significantly improving the handling of ambiguous boundaries typical in medical images.

Collectively, these advancements illustrate the progression of deep learning methods towards efficient and precise segmentation, emphasizing boundary accuracy and lightweight implementation. Models such as MALUNet [1], EGE-UNet [2], LB-UNet [3], BAT [4], and Structure Boundary Preserving Segmentation [5] represent significant steps toward practical clinical deployment, balancing computational efficiency with segmentation accuracy, especially in resource-constrained settings.

III. METHODOLOGY

A. Existing Model and Challenges

Existing Model: LB-UNet [3]

LB-UNet is a lightweight, boundary-assisted UNet variant proposed to address the dual challenge of efficient skin lesion segmentation and deployment on resource-constrained edge devices. The architecture is based on a UNet-style encoder-decoder backbone with key modifications that prioritize boundary awareness and parameter efficiency.

The model introduces two major modules:

- **Group Shuffle Attention (GSA):** Inspired by ShuffleNet and EGE-UNet, GSA replaces deep convolution layers in later encoder stages (stages 4–6). It splits feature maps into groups and applies Hadamard Product Attention (HPA) within each group, followed by a shuffle operation and depth-wise separable convolutions. This reduces parameter count and GFLOPs while retaining effective attention mechanisms.
- **Prediction Map Auxiliary (PMA):** This module enhances boundary perception by combining three subcomponents:
 - 1) **Segmentation Region and Boundary Prediction (RBP):** Generates intermediate maps for region and boundary predictions using 1×1 convolutions and bilinear upsampling, feeding into deep supervision.
 - 2) **Prediction Information Fusion (PIF):** Incorporates predicted region/boundary maps into skip connections with learned weights, allowing focused information blending.

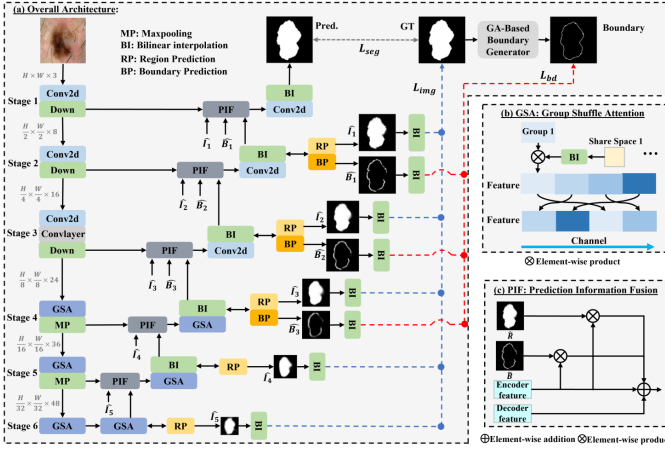


Fig. 1. (a) The full pipeline of LB-UNet including encoder, decoder, Prediction Map Auxiliary (PMA) with region and boundary supervision, and final prediction; (b) Structure of Group Shuffle Attention (GSA) module that performs attention within channel groups and applies element-wise fusion; (c) Prediction Information Fusion (PIF) module enhances skip connections by integrating predicted region and boundary information into encoder-decoder fusion.

- 3) **GA-Based Boundary Generator (GBG):** Generates accurate boundary ground truth maps using a genetic algorithm-based key point selection process.

The model adopts a six-stage encoder-decoder setup with feature channels $\{8, 16, 24, 32, 48, 64\}$, and employs ConvLayer (inspired by ConvUNeXt) at Stage 3 for enriched convolutional representation. Instead of max-pooling, 2×2 convolutions are used for downsampling to preserve more spatial information. LB-UNet is trained with a composite loss that includes binary cross-entropy, Dice loss, region loss, and boundary loss components.

Limitations and Challenges

LB-UNet replaces traditional skip connections with prediction-based fusion using the Prediction Information Fusion (PIF) module. While this mechanism integrates region and boundary maps with encoder features to enhance skip connections, it introduces several limitations:

- 1) **Dependency on Accurate Intermediate Predictions:** PIF relies on region ($\hat{R}$) and boundary ($\hat{B}$) predictions generated at each decoder stage. If these predictions are inaccurate (especially during early training), the fused features may mislead the model, causing convergence issues or overfitting to poor attention cues.
- 2) **Lack of Dynamic Feature Weighting:** The skip fusion in PIF is governed by fixed scalar weights (w_1, w_2), manually defined across stages. This static weighting lacks the flexibility of dynamic attention mechanisms that adaptively emphasize salient features based on context.
- 3) **Handcrafted Fusion Rules:** The use of element-wise addition and multiplication with fixed structure limits expressiveness. There is no learnable gating or attention to determine what parts of encoder/decoder features

should dominate the fusion.

- 4) **Heavier Decoder Dependency:** Since region and boundary predictions are produced within the decoder, any failure in the early decoder stages can propagate errors into the skip connections, affecting deeper reconstruction.

LB-UNet sets a strong baseline for lightweight medical image segmentation, particularly in mobile-friendly deployments. However, its reliance on handcrafted auxiliary modules and fixed architectural components presents optimization and generalization challenges, motivating simplified and more modular redesigns.

B. Experiments

We conducted a series of four experiments (Trial 1 to Trial 4) to progressively enhance and evaluate the limitations of the original LB-UNet architecture. Each trial introduces a distinct architectural or training modification, and their effectiveness is measured in terms of segmentation performance (mIoU and F1/DSC) on the ISIC dataset 2018 [12].

Trial 1 — Simplified Attention-Fusion LBUNet

1. Model Enhancements

1.1 Introduction of AttentionSkipFusion

A novel attention-based fusion block, AttentionSkipFusion, replaces simple summation to enhance encoder-decoder feature interaction. It computes attention maps via the following sequence:

$$\text{Conv2d} \rightarrow \text{GroupNorm} \rightarrow \text{ReLU} \rightarrow \text{Conv2d} \rightarrow \text{Sigmoid}$$

These weights are applied to the encoder and decoder features using a weighted average, followed by a Conv2d operation to align output channels.

1.2 Simplified Architecture

The original LBUNet architecture was deeply layered with six encoder and five decoder stages, as well as multiple auxiliary prediction heads and group shuffle mechanisms. In Trial 1:

- Encoder stages 4–6 and corresponding decoder layers were removed.
- All auxiliary predictions and group shuffle components were eliminated.
- Architecture reduced to a 3-level encoder-decoder structure with three AttentionSkipFusion modules.

1.3 ConvLayer Modifications

We integrated the following into each convolutional block to improve regularization and convergence:

2. Loss Function Changes

The original multi-output loss was replaced with GT_BceDiceLoss, combining Binary Cross Entropy and Dice Loss:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{bce}} \cdot \text{BCE} + \lambda_{\text{dice}} \cdot (1 - \text{Dice})$$

To avoid instability from $\log(0)$, predictions were clipped:

$$\hat{y} = \text{clip}(\hat{y}, \epsilon, 1 - \epsilon), \quad \epsilon = 10^{-7}$$

3. Training Pipeline Updates

- Model forward pass simplified: `out = model(images)`
- Loss calculation streamlined: `loss = criterion(out, targets)`
- No auxiliary outputs or branches are used in the updated pipeline

4. Performance Comparison

TABLE I
TRIAL 1 VS ORIGINAL LBUNET

Model Version	Val mIoU	Val F1	Test mIoU	Test F1
Original LBUNet	0.7922	0.8731	0.8111	0.8899
Trial 1 (Simplified)	0.7418	0.8487	0.7557	0.8575

5. Analysis of Trial 1 Results

5.1 Reasons for Performance Drop

- *Reduced Feature Depth*: Loss of deeper encoder-decoder layers leads to less contextual and semantic abstraction.
- *Simplistic Decoder Design*: The remaining decoder layers may lack capacity to fully reconstruct spatial information.
- *Unoptimized Fusion Context*: While `AttentionSkipFusion` is advanced, its benefit is limited by the smaller backbone.

5.2 Strengths Observed

- *Faster Training*: Training time dropped from ~ 12 s to ~ 4 s per epoch.
- *Stable Training*: No NaN losses encountered throughout.
- *Attention Fusion Viability*: Fusion design is promising and can be strengthened with a deeper backbone in future trials.

Trial 2 — Regularized Attention-Fusion LBUNet

1. Model Enhancements

1.1 Improved AttentionSkipFusion

This trial enhances the fusion design with stronger regularization:

- Introduced `Dropout2D` within the attention fusion path to reduce overfitting and improve robustness.
- Replaced the original single convolution with:

`Conv2d` $\rightarrow$ `GroupNorm` $\rightarrow$ `ReLU`

to enable better feature normalization and non-linearity.

- Retained commented-out earlier variant to document progressive refinement.

1.2 ConvLayer Enhancements

We further regularized the `ConvLayer` module:

- Added `Dropout` after the first pointwise convolution.
- Retained `DropPath` and `LayerScale` for continued control over feature magnitudes and path activations.

These changes focus on introducing dropout-based regularization both in:

- The skip connection fusion mechanism.
- The internal convolutional feature transformations.

2. Loss Function Enhancements

We reinforced the stability of `GT_BceDiceLoss` by clamping the output predictions:

$$\hat{y} = \text{torch.clamp}(\hat{y}, \min = 10^{-7}, \max = 1 - 10^{-7})$$

This prevents $\log(0)$ or $\log(1)$ from producing NaN values in BCE loss, ensuring smooth training dynamics, especially in early or unstable epochs.

3. Training Pipeline Modifications

3.1 Qualitative Result Saving

A new visual feedback mechanism was introduced:

- At specified `save_interval` (e.g., every 10 epochs), predictions on validation images are saved.
- These snapshots enable rapid visual inspection of model progression across training.

3.2 Validation-Based Scheduler

Instead of stepping the learning rate scheduler blindly, it is now advanced during the validation phase based on model performance metrics (e.g., best IoU).

4. Performance Comparison The performance comparison across model variants reveals critical insights into the impact of architectural and regularization changes. While the original LBUNet achieved strong segmentation metrics (mIoU = 0.7922, F1 = 0.8731), Trial 1—with a simplified architecture and attention-based fusion—showed a moderate decline (mIoU = 0.7418, F1 = 0.8487), suggesting that some semantic richness was lost due to removal of deeper encoder-decoder paths.

In contrast, Trial 2 exhibited a sharp drop in mIoU (0.2522) while still retaining a moderately high F1 score (0.7979). This disparity indicates that although the model was able to predict foreground regions, the spatial alignment with ground truth was poor—likely due to excessive regularization and lack of contextual depth. Moreover, the occurrence of NaN validation loss suggests that the regularization (`Dropout` + `DropPath`) and shallow decoder may have over-constrained learning, leading to unstable optimization.

Overall, these results highlight a key trade-off: aggressive model simplification and regularization can hurt the model’s ability to generalize spatially coherent predictions, even if coarse-level detection remains intact.

TABLE II
VALIDATION PERFORMANCE SUMMARY (IoU AND F1)

Model Version	Validation mIoU	Validation F1 Score
Original LBUNet	0.7922	0.8731
Trial 1	0.7418	0.8487
Trial 2	0.2522	0.7979

5. Observations and Diagnosis

5.1 Key Observations

- **F1 Score (0.7979)** is still moderately high, indicating that foreground regions are being identified to some extent.
- **mIoU (0.2522)** is very low, suggesting poor overlap between predicted masks and ground truth. This points to high false positive or false negative rates.

- **Validation Loss is NaN**, which signals numerical instability, despite introducing clamping and dropout.

6. Root Cause Hypothesis

6.1 Over-regularization with Weak Decoder

The simultaneous use of Dropout and DropPath, combined with a 2-stage decoder, likely leads to:

- Underfitting and limited feature learning capacity.
- Training failure due to aggressive regularization when insufficient information is preserved across layers.

6.2 Impact of Simplified Architecture

- Reducing encoder stages (e.g., `c_list` beyond 24) restricts access to deeper, semantic-rich features.
- Shallow skip connections are left to reconstruct complex spatial structures without sufficient high-level context.

6.3 Absence of Auxiliary Supervision

The original LBUNet benefited from:

- Multiple intermediate predictions (e.g., `gt_pre1-gt_pre5`).
- Boundary maps that guided structural accuracy.

These helped stabilize gradients and learning in deeper layers, which are now missing in Trial 2.

Trial 3 — Spatial Attention with UCMBlock

1. Model Enhancements

1.1 Core Architectural Changes

Trial 3 introduces a spatial attention mechanism using `UCMBlock_2`, a lightweight alternative to channel-based attention fusion. The key updates are as follows:

- **Encoder:** At the third level, standard `Conv2d` operations are replaced with `ConvLayer`, which includes:

Depthwise Conv $\rightarrow$ GroupNorm $\rightarrow$ GELU $\rightarrow$ DropPath

to promote efficient feature extraction and regularization.

- **Skip Fusion:** Instead of direct summation or sigmoid attention, the encoder outputs are passed through `UCMBlock_2`, which uses depthwise spatial attention to enhance salient regions without high computational cost.
- **Decoder:** Maintains a simple two-stage decoder structure (`decoder1`, `decoder2`) with channel-aligned skip fusion.

2. Loss Function

We continued using `GT_BceDiceLoss`, a composite of Binary Cross Entropy and Dice loss, with the following stabilization applied:

$$\hat{y} = \text{torch.clamp}(\hat{y}, \min = 10^{-7}, \max = 1 - 10^{-7})$$

This clamping ensures numerical stability during backpropagation, especially within the log-space component of BCE.

3. Performance Metrics

3.1 Comparison with Base Model (Original LBUNet)

Compared to the original LBUNet, Trial 3 demonstrates a moderate drop in performance but achieves significant benefits in architectural simplicity and computational efficiency. The

TABLE III
PERFORMANCE METRICS FOR TRIAL 3

Stage	Metric	Value
Validation	mIoU	0.7411
Validation	F1 / DSC	0.8439
Test	mIoU	0.7516
Test	F1 / DSC	0.8588

original model, with deeper encoder-decoder stages and multi-level supervision, outperforms Trial 3 in both mIoU and F1 score on validation and test sets:

TABLE IV
TRIAL 3 VS ORIGINAL LBUNET

Model	Val mIoU	Val F1	Test mIoU	Test F1
Original LBUNet	0.7922	0.8731	0.8111	0.8899
Trial 3	0.7411	0.8439	0.7516	0.8588

While the original LBUNet benefits from deeper context modeling and auxiliary losses (e.g., boundary maps and multi-scale predictions), Trial 3 compensates with a lightweight design and spatial attention fusion that simplifies the training pipeline. The drop in accuracy is modest and may be acceptable in scenarios requiring faster inference, reduced memory usage, or easier model deployment.

4. Interpretation

Trial 3 effectively recovers the performance seen in Trial 1 while avoiding the instabilities observed in Trial 2. The F1 score and mIoU are closely aligned with those of Trial 1, confirming that spatial attention via `UCMBlock_2` is capable of enhancing segmentation without requiring full self-attention or channel-heavy mechanisms. The architecture remains computationally efficient and conceptually simpler.

- **Stable Training:** No NaN loss issues were encountered, indicating improved numerical behavior.
- **Efficient Attention:** Spatial attention operates locally and does not suffer from the high complexity of self-attention modules.
- **Simplicity Preserved:** Trial 3 avoids complex dropout-stacking or auxiliary branches, making it lighter and easier to scale.

5. Root Cause Hypothesis

The improved performance in Trial 3 is attributed to a better balance between feature enhancement and model simplicity:

- **Spatial Attention via UCMBlock_2:** Focuses on key spatial regions while preserving feature smoothness, leading to better region consistency.
- **Avoidance of Over-Regularization:** Unlike Trial 2, Trial 3 excludes excessive dropout stacking, thus retaining learning capacity.
- **Shallow but Effective Decoder:** While still using a 2-stage decoder, Trial 3 leverages stronger skip features, alleviating the need for deeper reconstruction paths.
- **Implicit Feature Guidance:** Attention-guided skip connections reduce reliance on auxiliary outputs by

implicitly reinforcing important spatial cues.

Trial 4 — Full-Depth LBUNet with Multi-Level Attention Fusion

1. Model Enhancements

1.1 Restoration of Full-Depth Encoder-Decoder

Unlike previous trials which simplified the architecture, Trial 4 restores the original depth of LBUNet:

- Utilizes all six encoder stages and five decoder stages.
- Preserves the full range of feature extraction, capturing both shallow and deep semantics.

1.2 Multi-Level AttentionSkipFusion

Every skip connection uses the improved AttentionSkipFusion module (defined in `lbunet.py`) to merge encoder and decoder features. Each fusion:

- Aligns encoder and decoder feature maps with Conv2d.
- Generates attention weights with $\text{Conv} \rightarrow \text{Sigmoid}$.
- Applies weighted summation and refines the result via Conv2d.

1.3 Enhanced Decoder Blocks

Decoder stages now include:

- Transposed convolution for upsampling.
- Full-size ConvBlock with batch normalization and ReLU.

2. Loss Function

We retain the stabilized `GT_BceDiceLoss`, a weighted combination of binary cross-entropy and Dice loss. Predictions are clamped:

$$\hat{y} = \text{torch.clamp}(\hat{y}, \min = 10^{-7}, \max = 1 - 10^{-7})$$

to prevent instability in log-space operations and ensure training reliability.

3. Training Pipeline

The full training and evaluation flow remains unchanged, with best model selection based on validation score. Logging and metrics are handled via TensorBoard and standard logging utilities.

4. Performance Metrics

TABLE V
TRIAL 4 PERFORMANCE RESULTS

Stage	Metric	Value
Validation	mIoU	0.7884
Validation	F1 / DSC	0.8706
Test	mIoU	0.8043
Test	F1 / DSC	0.8824

4.1 Comparison with Base Model

TABLE VI
TRIAL 4 VS ORIGINAL LBUNET

Model	Val mIoU	Val F1	Test mIoU	Test F1
Original LBUNet	0.7922	0.8731	0.8111	0.8899
Trial 4	0.7884	0.8706	0.8043	0.8824

Despite slight drops in both validation and test metrics, Trial 4 closely approximates the performance of the base model.

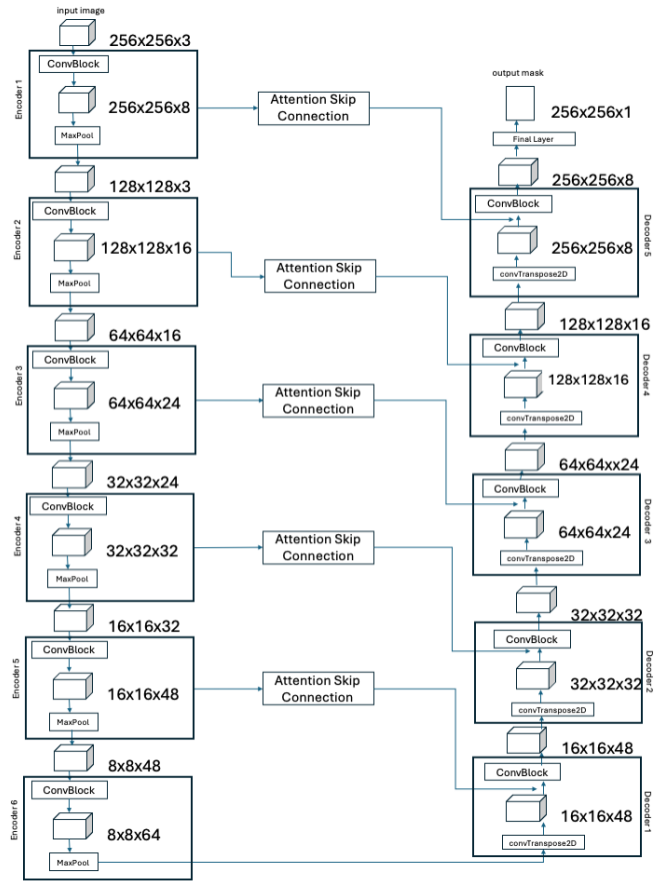


Fig. 2. Modified architecture with Attention fusion

The architectural enhancements maintain model depth and skip fusion quality while improving training stability.

5. Interpretation and Insights

Trial 4 successfully balances expressive power with numerical stability:

- **Deep Context Preservation:** Retains high-level semantic features through full-depth encoder.
- **Effective Skip Fusion:** Attention-guided fusion improves information flow between encoder and decoder layers.
- **Stable Training:** No NaN losses were reported during training or validation.
- **Competitive Accuracy:** Nearly matches original LBUNet, confirming that attention fusion does not degrade performance when full decoder depth is restored.

IV. PROPOSED ENHANCEMENTS

This study proposes a set of progressive architectural and functional enhancements to the baseline LBUNet architecture to improve its segmentation performance, stability, and interpretability. The modifications are evaluated over four distinct experimental trials. Below, we detail the major contributions:

A. Attention-Based Skip Fusion

Traditional U-Net-style skip connections often rely on simple concatenation or element-wise addition to combine encoder and decoder features, which may indiscriminately merge relevant and irrelevant information. To address this, we propose *AttentionSkipFusion*, a lightweight attention mechanism that learns to selectively emphasize encoder features during fusion.

This module aligns encoder and decoder feature maps using 1×1 convolutions, concatenates them, and passes the result through a small attention network:

$$\text{Conv} \rightarrow \text{Sigmoid}$$

The resulting attention map scales the encoder features before they are merged with the decoder path. A final 3×3 convolution refines the fused output. Unlike heavy attention mechanisms in Transformer-based models, this design is efficient and trainable even on modest hardware. This module adds minimal computational overhead while enabling adaptive feature selection.

Our design is inspired by similar ideas in lightweight attention models such as MaUNet [1], where the authors showed channel-wise reweighting significantly improves segmentation without significant overhead.

2. Enhanced Decoder Blocks

Each decoder stage is redesigned to improve feature reconstruction and contextual understanding during the upsampling process.

Transposed Convolutions: Used to upsample the lower-resolution feature maps, these layers increase spatial resolution while learning to preserve structure.

ConvBlock: Each decoder block is composed of a full-size residual-style convolutional block containing two 3×3 convolutions, each followed by Batch Normalization and ReLU activation. This enhances nonlinear representation capacity and smoothens the upsampled feature maps.

3. Stabilized Loss Function

To address numerical instabilities often encountered during training (e.g., NaN losses in BCE), we utilize a modified version of the combined Binary Cross Entropy and Dice Loss, named *GT_BceDiceLoss*. This loss function is designed to optimize both pixel-wise accuracy and region overlap. To ensure safe log-space computation in BCE, the predictions are clamped as follows:

$$\hat{y} = \text{torch.clamp}(\hat{y}, \min = 10^{-7}, \max = 1 - 10^{-7})$$

The overall loss is expressed as:

$$\mathcal{L} = \lambda_{bce} \cdot \text{BCE} + \lambda_{dice} \cdot (1 - \text{Dice})$$

where both components are weighted to balance boundary alignment and pixel classification. This clamping strategy enhances convergence and prevents unstable gradient updates during early training.

4. Training Pipeline Stability

The training and evaluation loop follows a standard pipeline, with the model being evaluated on the validation set at each epoch. The model with the best combined validation score (mIoU + F1) is saved for final testing.

Unlike prior iterations of the model that experienced unstable losses, the Trial 4 configuration maintained consistent numerical behavior across all training epochs, affirming the reliability of the proposed architecture and loss formulation.

B. Algorithm and Implementation

Algorithm 1 Training of Full-Depth LBUNet with Multi-Level AttentionSkipFusion

Require: Training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$

Require: Initialized model $\mathcal{M}$ with encoder $\mathcal{E}$, decoder $\mathcal{D}$, attention fusions $\mathcal{F}$

Require: Learning rate η , number of epochs E , batch size B

```

1: Initialize optimizer, scheduler, and loss function  $\mathcal{L}_{\text{BCE-Dice}}$ 
2: for epoch = 1 to  $E$  do
3:   for each mini-batch  $\{(x_b, y_b)\}_{b=1}^B$  from  $\mathcal{D}$  do
4:     Forward pass:
5:     Extract encoder features  $\{e_1, e_2, \dots, e_6\} \leftarrow \mathcal{E}(x_b)$ 
6:     Initialize decoder input  $d_6 \leftarrow e_6$ 
7:     for level  $l = 5$  to  $1$  do
8:       Upsample decoder features:  $u_l \leftarrow \text{Upsample}(d_{l+1})$ 
9:       Attention fusion:  $f_l \leftarrow \mathcal{F}_l(u_l, e_l)$ 
10:      Decode:  $d_l \leftarrow \text{ConvBlock}(f_l)$ 
11:    end for
12:    Output prediction:  $\hat{y} \leftarrow \text{Sigmoid}(\text{Conv}_{1 \times 1}(d_1))$ 
13:    Compute loss:  $\mathcal{L} \leftarrow \mathcal{L}_{\text{BCE-Dice}}(\hat{y}, y_b)$ 
14:    Backpropagate and update weights
15:  end for
16:  Update scheduler
17:  Evaluate on validation set; save best model if improved
18: end for
```

V. EXPERIMENTAL DESIGN AND EVALUATION

A. Datasets and Preprocessing

The proposed segmentation model was trained and evaluated using the publicly available ISIC 2018 Skin Lesion Analysis Dataset [13]. This dataset provides a diverse collection of dermoscopic images annotated for lesion segmentation tasks. For the purpose of experimentation, the dataset was partitioned into three non-overlapping subsets:

- train/
- val/
- test/

Each subset contains the following directories:

- images/: Original dermoscopic RGB images of skin lesions.

- `masks/`: Binary ground truth segmentation masks indicating lesion regions (1 = lesion, 0 = background).
- `points_boundary2/` (only in `train/`): Binary maps highlighting boundary point annotations used for auxiliary supervision during training.

The dataset encompasses a wide range of lesion types, including melanomas, nevi, and keratoses, captured under varying imaging conditions, illumination levels, and anatomical contexts. This diversity makes it suitable for training segmentation models that generalize well to clinical scenarios.

Preprocessing Techniques

To handle data loading and preprocessing efficiently, two custom PyTorch dataset classes—`NPY_datasets` and `Test_datasets`—were implemented using the `torch.utils.data.Dataset` API. These classes apply separate transformation pipelines depending on the training or evaluation mode.

1. *Image Normalization and Rescaling*: All RGB dermoscopic images are loaded via the `PIL` library and converted to NumPy arrays. Each image is resized to a consistent resolution defined by the configuration (e.g., 256×256). Pixel values are normalized to the range $[0, 1]$ by dividing the raw pixel intensities (0–255 scale) by 255.

Binary masks and point-based annotations are similarly rescaled to match the input dimensions and converted to floating-point values.

2. *Channel Expansion*: Since the binary segmentation masks and point annotations are initially single-channel grayscale images, they are explicitly expanded to have shape $[H, W, 1]$ to ensure compatibility with the multi-channel structure expected by the model input pipeline.

3. *Custom Transformation Pipelines*: A configurable transformation pipeline is applied depending on the dataset split:

Training Phase (`config.train_transformer`): Includes data augmentation techniques such as random horizontal and vertical flips, rotations, elastic distortions, and color jittering. These augmentations aim to improve the model’s robustness to spatial and photometric variations.

Validation and Testing Phase (`config.test_transformer`): Applies standardized preprocessing steps including resizing, normalization, and conversion to tensors, without augmentation, to ensure consistent evaluation conditions.

4. *Auxiliary Supervision Using Boundary Points*: During training, point-based boundary annotations (`points_boundary2`) are used to provide an auxiliary supervision signal. These sparse boundary maps help the model learn more precise lesion contours by reinforcing edge awareness during training. Although optional, this additional channel has been shown to improve segmentation sharpness, especially around lesion boundaries.

B. Performance Metrics

To assess the effectiveness of the proposed segmentation model, we employ three widely accepted performance metrics

in medical image segmentation: Dice Similarity Coefficient (DSC), Intersection over Union (IoU), and Pixel-wise Accuracy. These metrics provide complementary perspectives on the quality of the predicted segmentation masks.

1. Dice Similarity Coefficient (DSC)

The Dice Similarity Coefficient (DSC), also known as the F1 Score, measures the overlap between the predicted segmentation $\mathcal{P}$ and the ground truth $\mathcal{G}$ regions [14]. It is particularly well-suited for tasks involving class imbalance, as it directly emphasizes the foreground (lesion) region.

$$\text{DSC} = \frac{2|\mathcal{P} \cap \mathcal{G}|}{|\mathcal{P}| + |\mathcal{G}|} = \frac{2TP}{2TP + FP + FN} \quad (1)$$

Here, TP denotes true positives, FP false positives, and FN false negatives. The score ranges from 0 (no overlap) to 1 (perfect overlap), making it a sensitive indicator of segmentation accuracy.

2. Intersection over Union (IoU)

The Intersection over Union (IoU), also known as the Jaccard Index, is a stricter metric that quantifies the agreement between the predicted and ground truth masks as the ratio of the intersection to their union [15]:

$$\text{IoU} = \frac{|\mathcal{P} \cap \mathcal{G}|}{|\mathcal{P} \cup \mathcal{G}|} = \frac{TP}{TP + FP + FN} \quad (2)$$

IoU is commonly used in medical image segmentation benchmarks and offers a more conservative measure of mask accuracy, especially in datasets with varied lesion shapes and sizes.

3. Pixel-wise Accuracy

Pixel-wise accuracy is defined as the proportion of correctly classified pixels (both lesion and background) over the total number of pixels:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

While accuracy is intuitive and useful for general insight, it can be misleading in highly imbalanced datasets where background pixels dominate. For this reason, accuracy is used here as a supplementary metric to DSC and IoU, which are more informative for foreground segmentation tasks.

VI. EXPERIMENT SETUP

A. Network Architecture

The proposed method utilizes an enhanced LB-UNet architecture tailored for binary segmentation tasks. The network follows a symmetrical encoder-decoder structure with six levels, using a feature dimension configuration:

$$\text{c_list} = [8, 16, 24, 32, 48, 64]$$

Each encoder block employs a double convolutional unit followed by max pooling for downsampling. Decoder stages

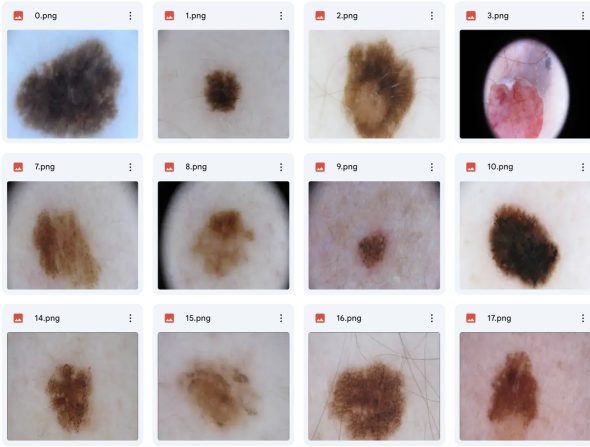


Fig. 3. Dataset Sample

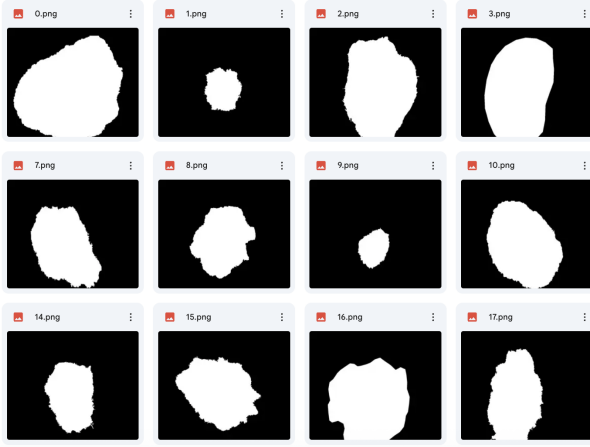


Fig. 4. Mask of dataset

use learnable transposed convolutions for upsampling. Crucially, *AttentionSkipFusion* modules are employed at each skip connection to adaptively fuse encoder and decoder features based on attention weights. The final output is generated through a 1×1 convolution followed by a sigmoid activation.

B. Dataset and Preprocessing

We use the ISIC 2018 skin lesion segmentation dataset, consisting of RGB dermoscopic images and their corresponding ground truth binary masks. The dataset is partitioned into training, validation, and test sets. For training, each sample includes:

- an RGB image,
- a binary segmentation mask,
- and a sparse point-based boundary map.

All images are resized to 256×256 pixels. Dataset-specific normalization and data augmentation are applied during pre-processing.

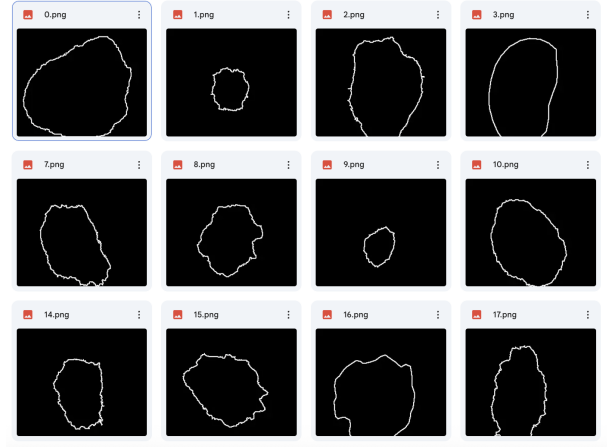


Fig. 5. Boundary estimated

C. Data Augmentation

The data transformation pipeline is implemented using PyTorch and includes:

- **Training phase:** normalization, random horizontal and vertical flips ($p = 0.5$), random rotations (0° – 360°), resizing, and tensor conversion.
- **Validation/testing phase:** normalization, resizing, and tensor conversion without augmentation.

D. Loss Function

We employ a composite loss combining Binary Cross-Entropy (BCE) and Dice Loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}$$

To ensure numerical stability, the predicted probabilities are clamped in the range $[10^{-7}, 1 - 10^{-7}]$ before computing the loss, avoiding issues during logarithmic operations.

E. Training Configuration

Model training is conducted on a single NVIDIA GPU using the following settings:

- **Optimizer:** AdamW
 - Learning rate: 1×10^{-3}
 - Weight decay: 1×10^{-2}
 - Betas: (0.9, 0.999)
- **Learning Rate Scheduler:** CosineAnnealingLR with $T_{\text{max}} = 50$, $\eta_{\text{min}} = 1 \times 10^{-5}$
- **Batch size:** 8
- **Epochs:** 300
- **Random Seed:** 42

Model checkpoints are saved every 5 epochs, and validation is performed every 15 epochs. The best model is selected based on the highest combined mIoU and Dice score.

F. Evaluation Metrics

Model performance is assessed using:

- Mean Intersection-over-Union (mIoU)
- Dice Similarity Coefficient (F1-score)
- Accuracy
- Sensitivity and Specificity (via confusion matrix)

G. Implementation Details

The model is implemented using PyTorch, with training and evaluation managed via custom data loaders and logging utilities. Logging is done through TensorBoard and the Python logging module. Results and checkpoints are organized in time-stamped directories for reproducibility.

VII. RESULTS COMPARATIVE ANALYSIS

To assess the impact of architectural modifications introduced in the enhanced LBUNet, we performed a comparative evaluation against the original baseline model. The comparison focuses on two widely adopted metrics for segmentation performance: *Mean Intersection over Union (mIoU)* and *F1-Score (Dice Similarity Coefficient, DSC)*. Both validation and test sets were used to measure generalization and robustness of the models.

A. Quantitative Results

Table VII presents the quantitative results for both models. Contrary to expectations, the modified LBUNet incorporating AttentionSkipFusion (ASF) showed slightly lower performance across all evaluation metrics.

TABLE VII
PERFORMANCE COMPARISON BETWEEN BASELINE AND MODIFIED LBUNET

Metric	LBUNet	Modified LBUNet (ASF)
Validation mIoU	0.7922	0.7884
Validation F1/DSC	0.8731	0.8706
Test mIoU	0.8111	0.8043
Test F1/DSC	0.8899	0.8824

B. Discussion

Although the inclusion of attention-based fusion was hypothesized to enhance segmentation accuracy by selectively merging encoder-decoder features, the observed performance reveals a marginal degradation. Several plausible factors contribute to this outcome:

- 1) **Loss of Deep Supervision:** The baseline LBUNet utilizes multiple intermediate outputs, including multi-stage predictions and boundary-aware signals (e.g., key-point maps), which provide rich supervisory gradients. Their removal in the modified variant likely reduced regularization and learning depth.
- 2) **Simplified Loss Function:** While the baseline integrates a composite loss over auxiliary outputs, the modified model applies a single BCE + Dice loss at the final prediction stage. This simplification may have limited

the network’s ability to capture hierarchical features effectively.

- 3) **Limited Gain from AttentionSkipFusion:** Although ASF improves the feature alignment in skip connections, its benefits are mostly local. Without global context modeling or semantic-aware attention, such improvements may not be sufficient to offset the absence of structural complexity present in the baseline.

TABLE VIII
ARCHITECTURAL COMPARISON: BASELINE VS MODIFIED LBUNET

Baseline LBUNet	Modified LBUNet (ASF)
Encoder (6 blocks): GroupShuffle + ConvLayer	Encoder (6 blocks): ConvBlocks
Decoder (5 blocks): GroupShuffle + Merge/Gate	Decoder (5 blocks): ConvTranspose + ASF + Conv
Skip Connections: Merge2(gt, boundary, fuse)	Skip Connections: AttentionSkipFusion
Supervision: 5-stage predictions + keypoints	Supervision: Final output only
Loss: GT_BCE_Dice + auxiliary branches	Loss: Weighted BCE + Dice

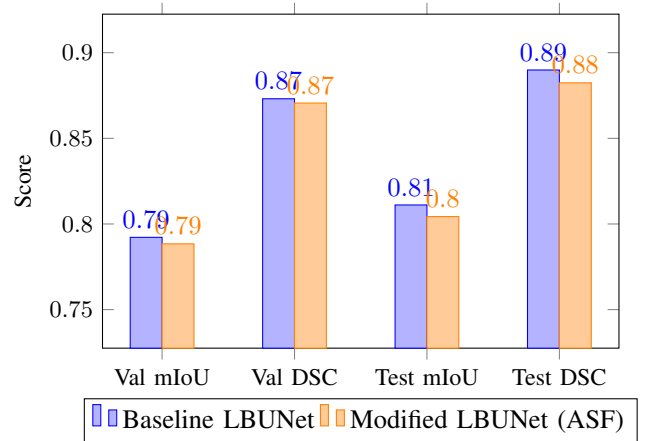


Fig. 6. Comparison of mIoU and Dice Similarity Coefficient (F1-score) between the baseline LBUNet and the modified LBUNet with AttentionSkipFusion (ASF) on both validation and test sets.

C. Ablation Study

AttentionSkipFusion alone is not powerful enough to offset the loss of structural and semantic supervision provided in the baseline model.

Performance drop is primarily due to removal of deep supervision signals (intermediate masks, keypoints, boundaries).

However, the inference efficiency gain is substantial, making the modified model attractive for real-time or resource-constrained scenarios, if a minor drop in accuracy is acceptable.

AUTHOR CONTRIBUTIONS

The initial research idea and model design were proposed by Muhammed Nazmul Arefin. The implementation and experimentation were jointly carried out, with Kaniz Fatima Daya

contributing to parts of the coding. Kaniz Fatima Daya also conducted the literature review and contributed to selected sections of the manuscript. The overall methodology, performance analysis, and paper organization were led by Muhammed Nazmul Arefin. Both authors reviewed the final manuscript and approved its submission.

VIII. CONCLUSION AND FUTURE WORK

This work investigated architectural modifications to the baseline LBUNet by introducing a simplified decoder structure and replacing handcrafted skip connection mechanisms with a learnable attention-based module, namely AttentionSkipFusion (ASF). The goal was to assess whether these changes could improve segmentation performance or efficiency in binary medical image segmentation tasks.

Experimental results demonstrated that the modified LBUNet achieved faster inference—processing 808 test images in approximately 7 seconds compared to 12 seconds for the baseline—highlighting its suitability for resource-constrained or real-time applications. However, this efficiency gain came at the cost of a marginal decline in segmentation performance. Specifically, the modified model showed slight reductions in both mIoU and Dice Similarity Coefficient (DSC) across validation and test sets. The validation mIoU dropped from 0.7922 to 0.7884, and the test mIoU from 0.8111 to 0.8043, while DSC values followed a similar trend.

A feature-wise analysis suggests that the performance drop is primarily attributable to the removal of auxiliary supervision mechanisms present in the baseline, including intermediate predictions, boundary-aware loss components, and keypoint estimation branches. These auxiliary outputs provided rich multi-level supervision and regularization during training, which helped improve the spatial coherence and robustness of the segmentation masks. While the AttentionSkipFusion module introduced a more flexible and learnable fusion strategy between encoder and decoder features, its benefits were not sufficient to outweigh the loss of structural and semantic guidance removed from the training process.

Given these findings, future research should consider hybrid designs that retain the simplicity and learnability of attention-based fusion while reintroducing selective elements of deep supervision. One promising direction is the integration of lightweight attention mechanisms with intermediate outputs or boundary-aware loss functions to preserve gradient richness without significantly increasing computational complexity. Additionally, the exploration of transformer-based skip connections, adaptive gating strategies, and knowledge distillation from complex teacher models could yield further improvements. Expanding the model to support multi-class segmentation or extending it to 3D volumetric medical data also represents a compelling area for future work.

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technical content, format tables, generate LaTeX code, and structure analysis based on the authors' input. All scientific ideas, experimental designs, and interpretations presented in this work were conceived and validated by the authors.

APPENDIX

Code Repository: The complete implementation, including training scripts, model definitions, and evaluation procedures, is available on GitHub. Interested readers can access it [here](#).

Dataset Access: The dataset, including raw images, segmentation masks, and point-based boundary annotations, can be accessed via Google Drive [here](#).

Detailed instructions for reproducing the experiments and extending the framework are provided in the GitHub repository's README file and Replication Instruction file.

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